

AMAN PARGANIHA

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Experience

Janyu Technologies

Mumbai, Maharashtra

Robotics Programming Intern

Aug 2026 – Present

- Building **MyFactory OS**, a manufacturing execution system for industrial-robotics production, combining an internal shop-floor job tracker with a customer-facing order-progress portal; owning backend delivery on a two-person team.
- Authored the platform's technical scope after benchmarking 12+ MES products (Siemens Opcenter, SAP ME, Tulip, Zetwerk), build-vs-buy case and MVP boundary approved by engineering leadership.
- Designed the relational schema and versioned REST API contract in **FastAPI**, **SQLAlchemy 2.0**, **PostgreSQL 16**, **Alembic** — JWT role-based access, query-level tenant scoping, transactional stage-advance logic with QC gating, and an append-only audit log.
- Containerised the stack with **Docker Compose** (API, PostgreSQL, MinIO, React/Vite frontend), keeping hosts and credentials environment-driven for on-premise deployment.

Education

International Institute of Information Technology Naya Raipur

India

M.Tech in Computer Science & Engineering (AI/ML) | CGPA: 8.0/10.0

Aug 2025 – Aug 2027

Shri Shankaracharya Technical Campus Bhilai

India

B.Tech in Computer Science & Engineering | CGPA: 8.22/10.0

Aug 2019 – Aug 2023

Technical Skills

Languages: Python, C++, TypeScript, SQL

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, diffusion models, vision-language models, reinforcement learning, NLP, model evaluation & benchmarking

LLMs & Generative AI: RAG pipelines, LlamaIndex, LangChain, OpenAI API (structured outputs, streaming), prompt engineering, embeddings, vector databases (Qdrant), agentic workflows, agentic coding tools (Claude Code)

Backend & MLOps: FastAPI, Node/Express, REST API design, server-sent events, SQLAlchemy, Alembic, Drizzle ORM, PostgreSQL, JWT auth, Docker Compose, AWS (EC2), MinIO/S3, Inngest, Git/GitHub, GitHub Actions CI

Data & Practices: Pandas, NumPy, ETL pipelines, dataset curation, feature engineering, SQL/NoSQL, unit & end-to-end testing (Vitest, Playwright), code review, system design

Projects

SayShip — Prompt-to-App Generation Platform | [Code](#) | [Live Demo](#)

TypeScript, React 19, Express 5, PostgreSQL, Drizzle, esbuild, OpenAI

- Built and deployed a platform turning a plain-English prompt into a running full-stack app: a planner agent emits a structured plan via **OpenAI Structured Outputs**, a writer agent streams files over **SSE**, an esbuild build check gates output, and a repair pass feeds compiler `file:line` errors back to the model.
- Executed untrusted generated code in an **opaque-origin sandbox** (iframe without `allow-same-origin` plus a CSP `sandbox` header) with a capability-keyed per-app CRUD API, so generated apps cannot reach platform sessions or data, added immutable versioning and an atomic DB lease for one run per project.
- Shipped with **67 Vitest/supertest** and **10 Playwright E2E tests** running against the production build in GitHub Actions CI, plus a mock LLM provider that makes the full product and test suite non-deterministically without an API key.

Event-Driven RAG Agent for Document Ingestion & Q&A | [Code](#) | [Live Demo](#)

Python, FastAPI, Inngest, LlamaIndex, Qdrant, OpenAI, Docker

- Built an event-driven RAG system (FastAPI backend, Streamlit chat UI, Dockerized Qdrant) that asynchronously ingests multi-page PDFs and answers queries with grounded, citation-backed responses.
- Architected durable, fault-tolerant ingestion with Inngest (independently retryable parse → embed → store steps) over LlamaIndex chunking, OpenAI embeddings and a Qdrant vector store, kept modular across chunking/embedding strategies.

Multimodal Credit Risk Analysis using SEC XBRL & NLP | [Code](#) | [Live Demo](#)

Python, Pandas, Scikit-learn, NLTK/VADER

- Engineered an end-to-end pipeline parsing ~41M raw SEC XBRL records (2022–2024) into a curated dataset of ~35,000 companies with 47 features (34 financial + 13 NLP-derived).
- Benchmarked Random Forest, Gradient Boosting, Logistic Regression and SVM, reaching **97.9% binary** / **93.8% multi-class accuracy** on financial features; ran per-feature-set evaluation and data-leakage checks to invalidate an inflated multimodal result.

Research

LiveJEPA: Real-Time Vision-Language Captioning (M.Tech thesis, in progress | PyTorch) — reconstructing Meta FAIR's VL-JEPA pipeline for streaming video captioning: frozen V-JEPA 2 encoder with a lightweight predictor head aligned to the SONAR text-embedding space, evaluated on EgoExo4D.

DDPM from Scratch: Denoising Diffusion Probabilistic Model ([Code](#) | PyTorch) — reimplemented DDPM from the paper without diffusion libraries: noise schedules, closed-form forward diffusion, UNet denoiser with sinusoidal timestep embeddings, and ϵ -prediction training with ancestral sampling, trained on Fashion-MNIST.

GreenCloudRL: Meta-RL for Energy-Efficient Cloud Scheduling (manuscript submitted) — hierarchical meta-RL scheduler (A2C/PPO with Reptile) with SHAP-based explainability, achieving ~14% energy reduction on Google and Alibaba cluster traces.

Open Source Contributions

Matplotlib — PR #31422 merged and shipped in v3.11.0: resolved issue #26620 by improving legend parameter documentation, reviewed and approved by core maintainer timhoffm.

JupyterLab — PR #18610 merged, PR #18668 approved: improved user-facing documentation and added Playwright end-to-end test coverage, both cleared multi-round maintainer review.

Certifications & Achievements

GATE CSE: Top 7.8% of ~210,000 candidates nationwide — secured M.Tech (AI/ML) at IIIT Naya Raipur. | **Problem Solving:** 200+ LeetCode problems solved. | **Google × Kaggle** — **5-Day AI Agents Intensive:** agentic LLM workflows, LlamaIndex/LangChain orchestration, RAG system design.